Paulsen's Lectures on Quantum Computing and Completely Positive Maps, Formalised in Isabelle/HOL

and a verified HHL algorithm

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October 2026

Abstract

We report on a formalisation in Isabelle/HOL of the mathematical content of Vern Paulsen's 2011 lecture course *Quantum Computing and Completely Positive Maps* (34 lectures): finite-dimensional Hilbert space theory, the postulates of quantum mechanics for pure and mixed states, partial traces, completely positive maps with Choi's theorem and the Choi–Kraus representation, the extreme points of the sets of unital and trace-preserving completely positive maps (Choi's Theorem 3), the Hahn–Banach theorem on matrix spaces, and the separating-hyperplane theory of entanglement witnesses. The development comprises forty-seven theories and about a thousand checked statements on top of the Jordan-Normal-Form and Isabelle Marries Dirac libraries; it builds without `sorry` or axioms and passes an oracle audit. A lecture-by-lecture audit records, for each statement in the notes, its formal counterpart or the reason it is absent, and an Isabelle-generated document reproduces the notes with the formal statements in place of the informal proofs. We then ask what this body of verified mathematics contributes towards a machine-checked proof of the Harrow–Hassidim–Lloyd (HHL) algorithm in Isabelle, and answer it with a sixth part beyond the notes: a functional calculus for Hermitian matrices, quantum phase estimation (exact case and the $4/\pi^2$ success bound, on the Archive's QFT circuit) and the HHL algorithm for eigenvalues representable in the register, with the output state, the solution property, a $1/\kappa^2$ success bound and the reduction of non-Hermitian systems. The for an eigenvalue that is not representable, the output of the physical (angle-clamped) circuit is within an explicit $O(1/(t_0\sqrt{N}))$ of the exact-HHL output, for every right-hand side.

1 Introduction

Machine-checked quantum computing in Isabelle/HOL has so far concentrated on algorithms: Isabelle Marries Dirac [2, 1] formalises qubits, gates, the no-cloning theorem, teleportation and the Deutsch and Deutsch–Jozsa algorithms on concrete matrices; the quantum Hoare logic of Liu et al. [9, 10] verifies Grover's algorithm; Manrique [11] adds the quantum Fourier transform. The mathematics that a graduate course on the subject spends most of its time on—positive matrices, the density-matrix formulation, partial traces, completely positive maps and their Choi matrices, the convex geometry of channels and of separable states—is covered unevenly. Unruh's libraries [3, 15, 16] develop much of this for arbitrary Hilbert spaces, in a type-based representation of operators; Echenim's entry [5] formalises projective measurements.

This paper describes a different kind of project: the end-to-end formalisation of one set of lecture notes, Paulsen's *Lectures on Quantum Computing and Completely Positive Maps* (Houston, Fall 2011) [12], taken as the specification. The notes are an unedited transcription of 34 lectures in five parts (basic mathematics; introduction to quantum computing; the theory of completely positive maps; entanglement; error correction). Following them lecture by lecture

forces one to formalise the whole supporting theory rather than the shortest path to a headline theorem, and it produces a natural, externally fixed measure of completeness: the audit of Section 4.

The project started as an autoformalisation experiment and was finished by the author with AI assistance; everything reported here is checked by Isabelle2025-2 with the strict build option `quick_and_dirty = false`, and the principal theorems have been checked to depend on no oracles. The sources, the audit, and the generated document are available in the project repository.[1]

Contributions. (1) A verified formalisation of Parts I–IV of the notes and of the elementary part of Part V: in particular Schur's theorem, the spectral theorem for Hermitian matrices, four characterisations of positivity, Cholesky's reduction, the measurement postulates and POVMs, the characterisation of density matrices by ensembles, partial traces and their compatibility with local measurements, Choi's theorem and the Choi–Kraus representation of channels, the adjoint duality between unital and trace-preserving maps, Choi's Theorem 3 on extreme points, the Hahn–Banach theorem for matrix spaces, and the entanglement-witness characterisation of separability. (2) A lecture-by-lecture audit that classifies every definition and result of the notes as formalised, formalised in a variant form, not formalised, or deliberately omitted. (3) A document, generated by Isabelle from the theories, that restates the notes with the formal statements replacing the informal proofs. (4) A sixth part beyond the notes: phase estimation and the HHL algorithm, verified on top of this material (Section 7).

2 The notes and the formalisation method

Representation. All objects of the notes are concrete: vectors in $\mathbb{C}^n$ and matrices in $M_{m,n}$. We use the Jordan-Normal-Form library [14]: vectors are complex Matrix.vec with a dimension hypothesis dim_vec v = n, matrices are complex Matrix.mat with carrier hypotheses A ∈ carrier_mat m n. Dimensions are hypotheses rather than types, which is how the notes treat them and which is the convention of Isabelle Marries Dirac, so that its gates, inner product, adjoint (dagger) and tensor products can be reused directly. Maps $\Phi : M_n \to M_d$ are HOL functions complex Matrix.mat ⇒ complex Matrix.mat constrained only on the carrier; linearity, positivity and complete positivity are predicates on such functions.

Two places in the notes genuinely need topology (compactness and the separation of a point from a closed convex cone, Days 22 and 27–28). For these we introduce a typed wrapper ('a,'m,'n) lmat of matrices of fixed dimension, instantiate the HOL-Analysis type classes up to euclidean_space and heine_borel, and transport the relevant statements across the wrapper. The carrier-predicate world and the typed world coexist; the former carries almost all of the development, the latter is used only where HOL-Analysis is applied.

Fidelity. Statements are formalised as close to the notes as the representation permits. The audit distinguishes faithful statements from *variants*; the main variants are: the predicate psd carries Hermiticity as an explicit conjunct (equivalent by the notes' own Theorem that $\langle x|Hx\rangle \in \mathbb{R}$ for all x iff $H = H^*$); Cholesky's reduction adds the hypothesis $p_{11} > 0$ that the notes leave implicit; linear maps are given with explicit additivity and homogeneity hypotheses on the carrier; and Choi's Theorem 3 is proved in its trace-preserving form with a fixed Kraus list of linearly independent operators, see Section 3. Where the notes themselves are inconsistent we formalise the correct statement and say so in the commentary (the notes claim $H^2 = X$ on Day 10 and correct it to $H^2 = I$ on Day 11; they use two sign conventions for the Pauli matrix Y).

[1] `https://github.com/brydustin/Quantum_Computing`

Verification discipline. The project builds as three Isabelle sessions with `quick_and_dirty = false`, so no `sorry` can survive. We also ran an oracle audit (Thm_Deps.all_oracles over the principal theorems). This turned up a methodological point worth recording: under `quick_and_dirty = true`, Isabelle's Goal.prove_sorry discharges package-internal goals of fun and record definitions with Skip_Proof.cheat_tac, so every theorem downstream of such a definition reports the oracle Pure.skip_proof even in a development that contains no `sorry` at all. Oracle audits are only meaningful on builds with the strict option.

3 What is formalised

Table 1 lists the theories. We describe the content by lecture group; the names in brackets are the Isabelle theorems.

Lectures 1–10: linear algebra and the postulates. Besides the inner-product algebra, matrix units and the trace (Tr_cyclic, hs_inner_prod_as_trace), the first part proves the projection formula $P = \sum |v_i\rangle\langle v_i|$ and Parseval (parseval_norm), the characterisations of unitaries, Householder unitaries and Schur's lemma (schur_lemma), Schur's triangularisation (schur_theorem) and $\operatorname{Tr} A = \sum \lambda_i$, the spectral theorem for Hermitian matrices (hermitian_spectral_decomp: $U^\dagger H U$ is a real diagonal matrix), and the positivity theory: $P \geq 0$ iff Hermitian with non-negative eigenvalues (psd_iff_nonneg_eigenvalues), iff $P = \sum |v_i\rangle\langle v_i|$ (psd_kraus_decomp, outer_sum_psd), positive definiteness iff the v_i span (pd_iff_spanning), the positive square root, and Cholesky's reduction (cholesky_reduction). Tensor products are Kronecker products with bilinearity, the multiplicativity of inner products and the orthonormal product basis (tensor_onb). The postulates are definitions: quantum_state, evolve, measurement_system, meas_prob, with the probabilities summing to one, POVMs with their square-root measurement systems (povm_from_sqrt), perfect distinguishability of orthogonal states, and local measurements in composite systems (composite_measurement_prob).

Lectures 11–22: channels and Choi's theorem. The no-cloning theorem (no_cloning) and the existence of oracles U_f (oracle_exists) open the second part. Ensembles and their density matrices are defined, and density matrices are characterised as the positive matrices of trace one (density_matrix_iff_ensemble, via the spectral ensemble); the ensembles of a pure state consist of multiples of the state (pure_ensemble_parallel). Partial traces are defined by the block formulas of the notes; they are trace preserving and positivity preserving (partial_trace_B_is_density) and compatible with local measurements: $\operatorname{Tr}((M^\dagger M \otimes I)\rho) = \operatorname{Tr}(M^\dagger M \operatorname{tr}_B \rho)$ (partial_trace_measurement). The coordinate form of $L(\mathcal{H}_A) \otimes L(\mathcal{H}_B) \cong L(\mathcal{H}_A \otimes \mathcal{H}_B)$ is an explicit linear isomorphism. Measurement maps $\rho \mapsto \sum M_\alpha \rho M_\alpha^\dagger$ preserve density matrices; Kraus maps are completely positive (kraus_is_cp), with complete positivity defined through the block amplification $\Phi^{(p)}$. Choi's theorem is choi_theorem: a linear Φ on the carrier is completely positive iff its Choi matrix $(\Phi(E_{ij}))_{ij}$ is positive, and kraus_rep_of_psd_choi extracts a Kraus representation with nd operators from a positive Choi matrix, which gives the Choi–Kraus characterisation of channels (choi_kraus_cptp) and the Kraus-rank bound (kraus_rank_bound). The Hilbert–Schmidt adjoint of a Kraus map is the Kraus map of the adjoint operators (kraus_adjoint_correct), and Φ is unital iff Φ^* is trace preserving (cp_unital_iff_adjoint_tp); positive functionals are completely positive (positive_scalar_is_cp); the sets UCP and CPTP are convex.

Lecture 23: Choi's Theorem 3. Choi's characterisation of the extreme points of UCP [4] is formalised in its trace-preserving form, for a channel given by a Kraus list $V_1, \ldots, V_r$ of linearly independent operators (choi_theorem_3_cptp_iff): Φ is extreme if and only if

$\{V_j^\dagger V_i\}$ is linearly independent over $\mathbb{C}$. (The perturbation argument of the notes produces a real-coefficient relation; a complex relation is first turned into a Hermitian one, using that the family $\{V_j^\dagger V_i\}$ is closed under adjoints, so both directions use the same notion of independence.) Two aspects deserve comment. First, because HOL functions are only constrained on the carrier, the naive definition of an extreme point ("Φ is not the midpoint of two *different* CPTP maps") is vacuous: any CPTP map agrees on the carrier with maps that differ from it elsewhere. Extremality is therefore defined through disagreement on the carrier (extreme_CPTP_carrier). Second, the notes' proof uses the unitary freedom of Kraus representations (their "Choi 2"), which was formalised only afterwards (choi_2_unitary_freedom in Day21_Unitary_Freedom). The forward direction instead goes through a CP Radon–Nikodym theorem (cp_radon_nikodym_linear): if Ψ and $\Phi - \Psi$ are completely positive and $\Phi(X) = \sum_i V_i X V_i^\dagger$, then $\Psi(X) = \sum_{ij} T_{ij} V_i X V_j^\dagger$ for a positive matrix T. Its proof reduces to a statement about positive matrices dominated by an outer sum, $0 \le B \le VV^\dagger$ implies $B = VTV^\dagger$ with $T \ge 0$, which we prove with the Moore–Penrose pseudoinverse of $VV^\dagger$ built from the spectral theorem: the identities $AA^+A = A$ and, for $0 \le B \le A$, $AA^+B = B$ follow from the kernel inclusion $\ker A \subseteq \ker B$, and $T = (A^+V)^\dagger B (A^+V)$.

Lectures 25–28: Hahn–Banach and entanglement witnesses. The notes state the Hahn–Banach theorem for bounded functionals on normed spaces and use it in Arveson's extension theorem. We prove it for matrix spaces with the Hilbert–Schmidt norm (matrix_hahn_banach_lmat): HOL's Hahn–Banach theorem gives a norm-preserving real-linear extension of the real part, and the Bohnenblust–Sobczyk formula $f(x) = F(x) - iF(ix)$ recovers a complex-linear extension with the same bound. Arveson's correspondence and extension theorem are formalised in the third block of Table 1 (Section 4). The entanglement theory of Days 27–28 is: the separable matrices form a closed convex cone (separable_cone_lmat_closed, using the closedness of the positive matrices, the compactness of the product density matrices and a Carathéodory bound on the number of generators), so a non-separable ρ is separated from the cone by a real functional, which becomes a Hermitian witness W with $\Re\,\mathrm{Tr}(W\rho) < 0 \le \Re\,\mathrm{Tr}(W\sigma)$ for all separable σ (entanglement_witness_hermitian'); this yields the Horodecki-type characterisation [8]: a Hermitian ρ is separable iff every such witness is non-negative on it (separable_iff_witness), with a strict version for density matrices. The notes phrase the criterion through positive maps, which needs the Arveson correspondence; the witness form is the equivalent statement that does not.

Lectures 29–32: error correction. Hamming weight and distance, the parity check and repetition codes, the Pauli matrices and their orthogonality, the bit-flip, phase-flip and bit-phase-flip channels (as UCP and CPTP maps), the pure-state fidelity, and the code words of the three-qubit codes are formalised. The recovery theorems for the three-qubit code and the Shor code, and the Knill–Laflamme conditions of the last lectures, are not.

4 The audit

The file `NOTES_AUDIT.md` in the repository lists, for each of the 34 lectures, every definition, theorem, proposition, corollary, example and remark of the notes with its formal counterpart. Each item is classified: ✓ faithful, ∼ formalised in a variant form, × not formalised, or − deliberately not formalised (overviews, history, numerical examples, classical asides such as stochastic matrices or cyclic codes). Of the 170 items, 10 are deliberate omissions; of the remaining 160 mathematical items, 153 are faithful, 7 are variants and none is a gap.[2] The

[2] Counts as of the third close-out of October 2026; the audit file is the authoritative record.

audit drove a final round of proofs: small statements omitted along the way (the converses isometry ⇒ unitary, positive definiteness iff positive eigenvalues, the expected value $\langle\psi|H\psi\rangle$ of an observable, the phase invariance of $|\psi\rangle\langle\psi|$, that $\rho_1 \otimes \rho_2$ is a density matrix, the unambiguous discrimination of linearly independent states, and the complex-coefficient form of Choi's Theorem 3) were proved during the close-out and are counted as faithful above. A second pass closed the medium-sized gaps that the notes prove in a page, each in its own theory (the second block of Table 1): a linear map sends density matrices to density matrices iff it is positive and trace preserving (density_preserving_iff_positive_tp); the transpose on M_2 is positive but not 2-positive, with the notes' witness (E_{ij}); the Hadamard power $H^{\otimes N}|0\cdots0\rangle = 2^{-N/2}\sum_J |J\rangle$ and the counting probability $p_0 = (M/2^N)^2$ of Day 12, for any unitary acting as the oracle on the basis; the Pauli expansion identity $\sum |a_k|^2 = 1$ for unitaries; the bound $\mathrm{cr}(\Phi) \le n$ for extreme CPTP maps (linear independence bounds the length, through the vector-space library of the Jordan-Normal-Form entry); the two-qubit Horodecki–Peres criterion, $(\mathrm{id}\otimes t)(|v\rangle\langle v|) \ge 0$ iff v is a product vector, with the witness value $-2|v_0v_3 - v_1v_2|^2$; and the three-qubit bit-flip code, whose syndrome projections, CPTP recovery map and recovery theorem $R(|\psi_i\rangle\langle\psi_i|) = |\psi\rangle\langle\psi|$ for at most one flip are proved on explicit 8×8 matrices. A third pass added the structural results of lectures 20–22 and 32–34: two rank-one decompositions $\sum v_j v_j^* = \sum w_l w_l^*$ with the v_j independent are related by an isometry, and hence two Kraus representations of one map by an isometry (Choi's Theorem 2, choi_2_unitary_freedom, proved through the Choi matrix); a Kraus representation of minimal length $\mathrm{cr}(\Phi)$ is linearly independent; the Choi rank theorem $\mathrm{cr}(\Phi) = \mathrm{rank}\, P_\Phi$ (choi_rank_eq_rank, with the rank taken in the Jordan-Normal-Form vector-space library); the general recovery proposition of Day 32, $R(|\varphi\rangle\langle\varphi|) = (\sum_i |a_i|^2)|\psi\rangle\langle\psi|$ for $\varphi = \sum_i a_i U_i\psi$ with $PU_i^*U_jP = \delta_{ij}P$ (recovery_linear_combination; the notes assume unitary U_i, but only $PU_i^*U_iP = P$ is used); and the Knill–Laflamme theory: the equivalence between correcting the states of V and fixing PXP, Theorem 1 in both directions (knill_laflamme; necessity through Choi's Theorem 2, sufficiency by diagonalising (α_{ij}) and applying the recovery proposition to the rescaled mixed errors, which need not be restrictions of unitaries) and the closure of the corrected errors under linear combinations. Finally the Shor code: one CPTP map corrects every single-qubit unitary error on each of the nine qubits (shor_code_theorem). The 2^9-dimensional space is never handled entrywise; nine qubits are three blocks of three, each single-qubit operator acts inside one block, inner products factor over the blocks, and the orthogonality of the 22 error images reduces to the distinctness of 44 block labels, decided by code_simp. Here the formalisation found a false statement: the Day 32 proposition that distinct 1-Paulis map the code space to orthogonal subspaces fails for $Z_1 \ne Z_2$, which agree on the code space (shor_notes_Z1_Z2); the notes correct it on Day 33. The environment model of Day 17 is proved as stated: $\mathrm{tr}_E(U(\rho\otimes|\varphi\rangle\langle\varphi|)U^*) = \sum_i W_i\rho W_i^*$ with $\sum_i W_i^*W_i = I$ (environment_model). Last, the operator-system part of the course: Arveson's theorem that a linear map on an operator system is completely positive iff it is d-positive iff S_Φ is positive, the correspondence $\Phi \leftrightarrow S_\Phi$, the extension theorem (arveson_extension) with the Kraus form of completely positive maps on operator systems, and the positive-map criterion for separability (separable_iff_positive_maps). The extension is not proved as in the notes (a state of norm one, Hahn–Banach, and a reduction to the unital case): the positive functional S_Φ on $M_d(S)$ is extended directly, by separating the positive definite cone from its kernel with HOL-Analysis' separating hyperplane theorem in the Euclidean space of typed matrices, which gives the extension $X \mapsto \mathrm{Tr}(DX)$ with $D \ge 0$. A fourth pass (the fourth block of Table 1) proved the examples and small propositions that remained: the projection does not depend on the orthonormal basis; the unique coordinates $u = \sum_j x_j \otimes f_j$ for any basis and $X\otimes Y \cong X\oplus\cdots\oplus X$; outcome i fires on ψ_2 with probability at least $|\langle\psi_1|\psi_2\rangle|^2$ if it is certain on ψ_1; the Bell-pair measurement of Day 9; SWAP = CNOT · CNOT′ · CNOT; the classical error probabilities of Day 30; the bit-flip code under independent flips, with $R\circ E$ and both minimum fidelities of Day 31; and the code for Y flips. Two statements of the notes needed correction. The Day 29

"Hamming $[7,1,3]$" code (seven data bits, three parity bits) has distance 2: it detects one error but cannot correct one (notes_hamming_not_correcting); the standard $[7,4,3]$ code corrects one. The Day 31 formula $R \circ E(Y) = c_1 Y + c_2 XYX$ holds for Y supported on the code space, not for a general Y. The 6 remaining gaps are Sylvester's criterion (stated without proof in the notes), the Choi-matrix form of trace preservation and unitality, two lists of examples, and the two lemmas that only the notes' route to the extension theorem uses.

Beyond the omissions, the audit documents where the formalisation strengthens the notes (strict witnesses for density matrices; Cholesky with the missing hypothesis; the vacuity of the naive extreme-point definition) and where it changes the route of a proof (Choi's Theorem 3 through CP Radon–Nikodym).

A fifth pass (the fifth block) proved the three items that remained: the Choi-matrix form of trace preservation and unitality, the Day 23 examples, and the proposition that a unital functional is positive iff it has norm one. Two examples needed correction: the average $\frac{1}{2}(U_1^* X U_1 + U_2^* X U_2)$ is non-extreme only when U_1, U_2 are linearly independent (it is extreme when U_2 is a unimodular multiple of U_1), and the map $X \mapsto \mathrm{Tr}(X) I_d$ is trace preserving only after division by d. The norm proposition is proved without the spectral theorem, from $P^2 \leq \|P\| P$ for $P \geq 0$.

A sixth pass (the sixth block) closed the last three gaps. The two Day 26 factorisations of a completely positive map, $\Phi(X) = V^* \psi(X) V$ with ψ unital and $\Phi(X) = \psi(VXV^*)$ with ψ trace preserving (cp_unital_factor, cp_tp_factor), are proved from the Kraus form on the operator system with block embeddings B_i satisfying $\sum_i B_i B_i^* = I$, instead of the notes' compression to the range of $\Phi(I)$ and dualisation. The Day 27 examples $\left(\begin{smallmatrix} E_{ii} & E_{ij} \\ E_{ji} & E_{jj} \end{smallmatrix}\right)$ and (E_{ij}) are positive and not separable, since the transpose's ampliation takes the value -2 on $e_0 \otimes e_j - e_1 \otimes e_i$ (entangled_example_2, entangled_example_choi_seed). Sylvester's criterion is not reproved: it is imported from the AFP entry *Two Hermitian Results*, whose notions of Hermitian and positive definite matrices are shown to agree with ours (pd_iff_leading_minors_pos). Because that entry builds on HOL-Analysis, HOL-Probability and QHLProver, the bridge is a separate session with its own document; the main document states the criterion in its Day 5 commentary. The two numerical 3×3 examples of Cholesky's algorithm on Day 5, first counted as illustrations, are also checked step by step (notes_example_1, notes_example_2), together with the rank-one decomposition the algorithm produces. Seven variant rows were completed: the superdense-coding table (the notes' fourth matrix is ZX), the CCNOT truth table, and the partial trace over A, obtained from the B side by the index permutation that exchanges the factors (partial_trace_A_is_density); the converse directions of the Hermitian tests, the density of a composite ensemble, Choi's theorem with n-positivity as its own condition (choi_four_conditions), and the distance of subspace codes with unique decoding; and the ranges of positive matrices on Day 21 ($R(\sum_l w_l w_l^*) = \mathrm{span}\{w_l\}$, outer_sum_range), which also gives the Day 4 statement in its literal form: $\sum_i v_i v_i^* > 0$ iff the v_i span $\mathbb{C}^n$. The Schmidt rank was defined as the *minimum* of the set of lengths of representations by elementary tensors; that set is infinite, and Isabelle's Min is unspecified on infinite sets, so the definition was replaced by the least element. With it, a nonzero vector has Schmidt rank 1 iff it is a product state, and is entangled iff its Schmidt rank exceeds 1 (entangled_iff_schmidt_rank). Finally, bases of $\mathbb{C}^n$ and $\mathbb{C}^k$ give a basis of $\mathbb{C}^n \oplus \mathbb{C}^k$ (Day 5, direct_sum_basis_is_basis), and the adjoint Φ^* of a linear map makes $\Gamma(\Phi) = \Phi^*$ an affine bijection from the unital onto the trace-preserving completely positive maps (Day 22, adjoint_ucp_imp_cptp, map_adjoint_adjoint); through it, Choi's Theorem 3 is transported from the trace-preserving to the unital form stated in the notes (choi_theorem_3_ucp). The block picture of Day 16, $X = \sum_{i,j} X_{ij} \otimes E_{ij}$ with $H_A \otimes H_B \cong H_A^{(p)}$ isometric, is proved as well (tensor_block_decomposition), and so is the compactness of the unital completely positive maps, in Choi coordinates (compact_ucp). The seven remaining variants are deliberate: a different proof route (Choi's Theorem 3 through

CP Radon–Nikodym, the extension theorem by cone separation), a concrete model in place of an abstract one (the Kronecker tensor product, the coordinate isomorphism of Day 15), or a statement of the notes that needs a hypothesis or is false as stated (Cholesky with $p_{11} > 0$, the Day 23 examples, the Day 32 Pauli claim).

5 The generated document

Isabelle's document preparation turns the theories into a PDF in which proofs are folded and the commentary appears between the formal statements. During this project we rewrote the top-level comments of all theories as document markup: each lecture is a section, and before each group of formal statements a text block restates the notes' definitions and results in their own words and LaTeX, says which Isabelle statements correspond to them and how they differ, and names the audit items for what is missing. The result is a 2011 lecture course in which every displayed theorem has been checked, organised by theory rather than strictly by date because later lectures reuse earlier infrastructure; a reading guide at the front gives the map from lectures to theories. The document is produced by a fourth session, Vern_Paulsen_QC_Notes, through Isabelle's `document_theories` mechanism, so it is regenerated and re-checked by every build.

6 Relation to the Archive of Formal Proofs

Isabelle Marries Dirac [1] and the quantum Fourier transform entry [11] use the same matrix representation as this project (complex Matrix.mat with carriers), and we import both; our tensor-product lemmas (mixed product, adjoint, unitarity, product bases), measurement calculus and density-matrix theory extend them in the same style. The quantum Hoare logic entry [10] contains a positivity predicate, the Loewner order and a decomposition of positive matrices, overlapping with our Days 4 and 18, but no Choi theory. Unruh's *Complex Bounded Operators*, *Tensor Products in Hilbert Spaces* and *Kraus Maps* [3, 15, 16] are far more general—arbitrary Hilbert spaces, trace-class and Hilbert–Schmidt operators, the spectral theorem for compact operators, Kraus families for possibly infinite-dimensional channels—but live in a type-based representation of operators ('a $\Rightarrow_{CL}$ 'b); they contain neither Choi's theorem nor the extreme-point and separability results, and connecting them to the matrix world would require a transfer layer of the kind our lmat wrapper provides in miniature. Echenim's projective measurements [5] are formulated on complex^'n vectors. To our knowledge the results of Days 19–28 (Choi's theorem in its matrix form with explicit Kraus extraction, Choi's Theorem 3, the Hahn–Banach theorem on matrix spaces, the closedness of the separable cone and the witness characterisation) have no counterpart in the Archive.

7 A verified HHL algorithm

The HHL algorithm [7] prepares, for a Hermitian A with eigen-decomposition $A = \sum_j \lambda_j |u_j\rangle\langle u_j|$ and a unit vector $|b\rangle = \sum_j \beta_j |u_j\rangle$, a state proportional to $A^{-1}|b\rangle$: phase estimation with the unitaries e^{iAt} writes the eigenvalues into a register, a controlled rotation of an ancilla produces the amplitudes C/λ_j, uncomputation and a measurement of the ancilla postselect the branch $\sum_j (\beta_j/\lambda_j)|u_j\rangle$, up to the approximation error of phase estimation and with a success probability governed by the condition number. Machine checked proofs exist in Coq, in the quantum Hoare logic of Zhou, Yu and Ying [18, 17, 6]; none existed in Isabelle/HOL, whose verified algorithms were Deutsch–Jozsa, Grover and the QFT (a recent HOL Light development treats Grover [13]). Part VI of the development (session Vern_Paulsen_QC_Algorithms,

the last block of Table 1) builds HHL on the library of Parts I–V; it goes beyond the notes and is not counted in the audit.

Functional calculus. For Hermitian A and $g : \mathbb{R} \to \mathbb{C}$ we define $g(A)$ as the matrix with $g(A)v = g(\lambda)v$ for every eigenvector v with eigenvalue λ. The eigenvectors span, so this determines $g(A)$; no interpolation or independence-of-decomposition argument is needed. The spectral theorem of Part I gives existence and $g(A) = U\,\mathrm{diag}(g(D))\,U^\dagger$ for every decomposition. Multiplicativity, $g(A)^\dagger = \bar{g}(A)$, unitarity for $|g| = 1$ and positivity for $g \geq 0$ on the spectrum follow by checking eigenvectors. Instances are e^{itA}, with its group law, and A^{-1} (the pseudo-inverse when 0 is an eigenvalue, since $1/0 = 0$ in HOL).

Phase estimation. The Fourier matrix F_N, $(F_N)_{jk} = \omega^{jk}/\sqrt{N}$, is unitary by the roots-of-unity sum. Controlled powers $\sum_k |k\rangle\langle k| \otimes U^k$ act by phase kickback, and for $Uu = e^{2\pi i\varphi}u$ the circuit leaves the register in $\sum_j \alpha_j|j\rangle$ with $\alpha_j = \frac{1}{N}\sum_k e^{2\pi ik(\varphi - j/N)}$. If $\varphi = j_0/N$ the outcome is j_0. Otherwise the outcome nearest $N\varphi$ (mod N) has probability at least $4/\pi^2$, by Jordan's inequality and $|e^{ix} - 1| = 2|\sin(x/2)|$. The Archive's QFT entry [11] gives its circuit's output in product form; by induction over the Kronecker list we show that the circuit ordered_QFT n equals F_{2^n}, so these theorems hold for the Archive's circuit.

HHL. The state space is register $\otimes$ (system $\otimes$ ancilla), so phase estimation applies unchanged to $e^{it_0A} \otimes I$. The circuit is $W^\dagger\, R\, W$ with W phase estimation and $R = \sum_k |k\rangle\langle k| \otimes I \otimes R(C/\lambda_k)$; register value k stands for $\lambda_k = 2\pi k/(t_0N)$, or $2\pi(k-N)/(t_0N)$ for $k \geq N/2$. Assuming every eigenvalue of an invertible Hermitian A is some λ_k, we prove that the output on $|u\rangle \otimes |b\rangle \otimes |0\rangle$ (u the uniform register state) is

$$u \otimes \left(\sqrt{I - C^2A^{-2}}\, b \otimes |0\rangle + C\, A^{-1}b \otimes |1\rangle\right).$$

The proof treats an eigenvector and extends to all b by the fact that two matrices agreeing on all eigenvectors are equal, so no sums over an eigenbasis appear. Hence $x = A^{-1}b$ solves $Ax = b$, the ancilla outcome $|1\rangle$ has probability $C^2\|x\|^2$, and for $C \leq |\lambda| \leq \Lambda$ on the spectrum this is at least $(C/\Lambda)^2\|b\|^2$, i.e. $1/\kappa^2$ for unit b and $C = \lambda_{\min}$. A non-Hermitian invertible A is handled through $\left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)$, whose inverse maps $b \oplus 0$ to $0 \oplus x$.

Inexact phase estimation. When λ is not a register value, phase estimation leaves the register in $\sum_j \alpha_j|j\rangle$, the rotation acts with C/λ_j on the j-th branch and uncomputation is inexact. Writing states as register sums $\sum_j |j\rangle \otimes f_j$ (block operators act branch by branch, branches are orthogonal) gives the exact identity $\|\text{output} - u_N \otimes u \otimes R(C/\lambda)|0\rangle\|^2 = \|u\|^2 \sum_j |\alpha_j|^2 \|R(c_j)|0\rangle - R(c)|0\rangle\|^2$. For register values below C the rotation $R(C/\lambda_j)$ is not unitary and the branch error is unbounded, so the physical circuit clamps the angle to $[-1, 1]$; for it, with $\sum_j |\alpha_j|^2 = 1$, the tail bound $|\alpha_j| \leq 1/(2N|\delta_j|)$, Jordan's inequality and a Lipschitz estimate for $c \mapsto R(c)|0\rangle$ on $[0, \theta]$, we prove: if $\lambda \geq 2C/\theta$ and $t_0\lambda \leq \pi/2$ then the squared distance is at most $\|u\|^2\, \frac{\pi^2}{Nt_0^2}\left(\frac{\theta^4}{C^2(1-\theta^2)} + \frac{16}{\lambda^2}\right)$ (hhl_clamped_error_bound). The error vectors of distinct eigenvectors are orthogonal (each lies in $\mathbb{C}^N \otimes \mathrm{span}(u_j) \otimes \mathbb{C}^2$), so with U the eigenbasis, U^*E^*EU is diagonal for the error operator E and $\varepsilon^2 I - E^*E$ is positive: for every b, with the spectrum in $[\lambda_{\min}, \pi/2t_0]$ and $\lambda_{\min} \geq 2C/\theta$, $\|\text{output} - \text{exact output}\|^2 \leq \frac{\pi^2}{Nt_0^2}\left(\frac{\theta^4}{C^2(1-\theta^2)} + \frac{16}{\lambda_{\min}^2}\right)\|b\|^2$ (hhl_clamped_bound), with no loss in the dimension.

Sharper constants. The bound above is $O(1/N)$ in the squared error, so precision ε needs a register of size $N \sim 1/\varepsilon^2$. The uniform register state is the culprit: its amplitudes decay only like $1/(N\delta)$. Preparing the register in the sine window $\psi_\tau = \sqrt{2/N}\,\sin(\pi(\tau + \frac{1}{2})/N)$,

as in the HHL paper, the window sum has a closed form and $|\alpha_j|^2 \le \pi^2/(2N^4\delta_j^4)$ for $1/N \le |\delta_j| \le \frac{1}{2}$ (win_amp_tail). On good branches the distance is $O(\delta_j^2)$, so each term is at most $K\, w(N\varphi - j)/N^2$ for a weight w with $\sum_j w(N\varphi - j) \le 4 + 2\pi^2$; clamped and negative branches have $|\delta_j| \ge \varphi/2$ and contribute $O(1/N^3)$ in total. For every b, $\|\text{output} - \text{exact output}\|^2 \le \left(\frac{8\pi^2(2+\pi^2)\theta^4}{N^2 t_0^2 C^2(1-\theta^2)} + \frac{512\pi^6}{N^3 t_0^4 \lambda_{\min}^4}\right)\|b\|^2$ (hhl_window_bound), so $N \ge Q/\varepsilon$ gives error at most $\varepsilon\|b\|$ (hhl_window_precision).

The surrounding algorithm. The circuit uses e^{iAt_0} only through controlled powers, so we write it in a small language of fixed gates and controlled oracle calls with a semantics and a call count: HHL makes $2(N-1)$ calls, and replacing the oracle by any unitary V with $\|Vv - e^{iAt_0}v\| \le \eta\|v\|$ moves the output by at most $2(N-1)\eta$ (hhl_sim_robust, from a general lemma: each call adds at most η). End to end the output is within $(\varepsilon + 2(N-1)\eta)\|b\|$ of the exact one. The answer is read by post-selecting the ancilla; normalisation is stable, so the post-selected state is within $2\delta/(C\|A^{-1}b\|)$ of $\psi \otimes A^{-1}b/\|A^{-1}b\| \otimes |1\rangle$ (hhl_readout). Amplitude amplification is proved in the plane of the good and bad components (k rounds give $\sin^2((2k+1)\theta)$), and as a circuit it uses the HHL circuit and its inverse once per round. Theorem hhl_complete combines state preparation, the window, the simulation oracle, amplification and readout: success probability at least $1-p$ after $k \le \pi^2/(4Ct_0)$ rounds, i.e. $O(\kappa)$, with $(2k+1)\cdot 2(N-1)$ oracle calls.

The paper's error analysis. Appendix A of the HHL paper uses a three-level flag register (nothing, well, ill) set by filter functions f, g. We formalise f, g and $|h(\lambda)\rangle$, Lemma 2 ($\lambda \mapsto |h(\lambda)\rangle$ is $\frac{\pi}{2}\kappa$-Lipschitz; the constant is attained) and Lemma 3, the relative bound $|f-\tilde{f}|^2 + |g-\tilde{g}|^2 \le c\,\kappa^2(\lambda-\tilde{\lambda})^2(f^2+g^2)$. The paper states $c = \pi^2/2$; this is false. On $[1/2\kappa, 1/\kappa)$ the proof uses $f^2 + g^2 = \frac{1}{2}$, but there $f^2 + g^2 = \frac{1}{4}$, and the ratio of the two sides tends to π^2. We prove the lemma with $c = \pi^2$, which is sharp, by gluing region estimates (lemma3). With the window register and the wrapped distances of the register eigenvalues, all three parts of Theorem 1 follow for every b with explicit constants: $\|(\tilde{U} - U)b\| \le 3\pi^2\sqrt{4 + 2\pi^2}\,\kappa\|b\|/t$, and the post-selected states are within $12\pi^2\sqrt{4+2\pi^2}\,\kappa/t$ of the ideal ones, $t = Nt_0$ the total evolution time (flag_bound, flag_post_normalized, flag_well_normalized). The audit of the paper's results against the formalisation is part of the notes audit. The concentration bound (A32), $|\alpha_{k|j}| \le 8\pi/\delta^2$, holds for $2\pi \le |\delta| \le N$ (win_amp_A32). For the paper's run-time claim we work in its normalisation (spectrum in $[1/\kappa, 1]$): total evolution time $t \ge c\kappa/\varepsilon$ gives accuracy ε for the post-selected state, the success amplitude is at least $1/(4\kappa)$, and amplification takes at most $\pi\kappa$ rounds, so the oracle is run for total time $O(\kappa^2/\varepsilon)$ and the state preparation $O(\kappa)$ times (hhl_paper_runtime). Running amplification without knowing p rests on BHMT's averaging lemma (bhmt_quarter): stages with a random number of rounds below $1, 2, 4, \ldots$, stopped at the first success, succeed with probability at least $1/4$ in fewer than $8\sqrt{2}\,\kappa$ rounds (doubling_schedule_kappa). The reduction from circuits to matrix inversion behind the optimality results is formalised as a construction (red_inverse, red_A_bounds, red_clock_prob, red_post_state), with two corrections: the undo gates are U^*_{T+1-t}, not U^*_{3T+1-t}, and the condition number is $\coth\frac{1}{2T} \le 2T+1$ rather than at most $2T$. The complexity-class consequences need a theory of BQP, PSPACE and oracle lower bounds and are not formalised.

Rectangular systems and the Discussion. For $A \in \mathbb{C}^{M\times N}$ the algorithm runs on $H = \left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$ and inverts it on its non-zero eigenspaces. The pseudo-inverse maps $b \oplus 0$ to $0 \oplus x$ with $A^*Ax = A^*b$ and x in the range of A^*, so x is the minimum-norm solution when $Ax = b$ is solvable and a least-squares solution otherwise (pinv_solution, normal_eq_least_squares); the proof uses $Jg(H)J = g(-H)$ for the parity J. For Section IV we prove that rotation angle $h(\lambda)$ yields $h(A)|b\rangle$, that expectations, sampled distributions and k-fold copies move by

$O(\|\tilde{x} - x\|)$, and that the SWAP test, built from $H \otimes I$ and a controlled SWAP, reads $|0\rangle$ with probability $(1 + |\langle\psi|\varphi\rangle|^2)/2$. The estimation procedure of Appendix A5 is formalised with an expectation operator over n independent bits: mean, variance and Chebyshev give $3/\varepsilon^2$ samples (estimate_expectation). Two more errata: the paper's $1/(3\varepsilon^2)$ samples are too few (for $\varepsilon = \frac{1}{4}$, $p = \frac{1}{2}$ and six samples the estimate fails with probability $\frac{11}{16}$), and its preconditioned system $(AB)c = Bb$ is neither left nor right preconditioning; in a 2×2 example its unique solution gives no solution of $Ax = b$ (paper_precondition_wrong).

What remains. The window state and $|b\rangle$ are prepared by given unitaries. Hamiltonian simulation enters as an η-accurate oracle: building one from sparse access (Trotter-type formulas) and counting its gates is outside the matrix framework, so the complexity statement is in oracle calls.

8 Conclusion

Taking a complete set of lecture notes as the specification turned out to be a productive way to build a library: it fixes the scope, orders the work, and makes completeness measurable. The cost is visible in the audit: a course mentions many results it does not prove, and a formalisation faithful to the notes must decide, item by item, what to prove, what to restate, and what to leave. The commentary pass that produced the generated document is the record of those decisions; we recommend it as a deliverable for any formalisation of a text, because it is the only place where the informal and the formal statements are read side by side.

Acknowledgements. The notes were written by Vern Paulsen and transcribed by Brandon Lee. The formalisation was carried out with the assistance of AI coding agents; all statements were checked by Isabelle.

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